



جامعة الملك فهد للبترول والمعادن
King Fahd University of Petroleum & Minerals

Feedback Circuits (02-03)

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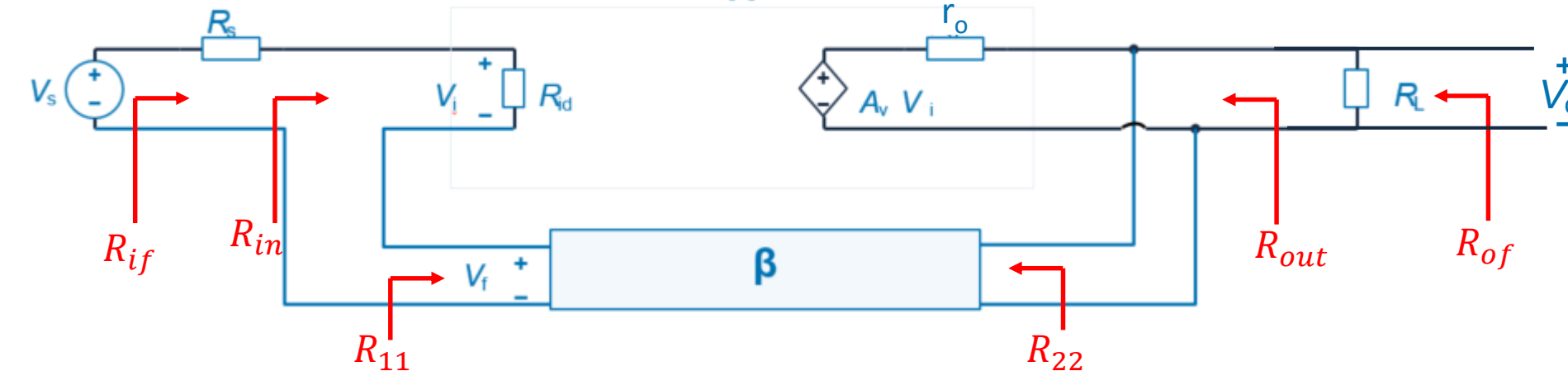


- ☐ **Practical Feedback Effect**
- ☐ **Steps for Practical Feedback Analysis**
 - ☐ **Series-Shunt Topology: Example 01**
 - ☐ **Series-Shunt Topology: Example 02**
 - ☐ **Series-Series Topology: Example 03**
 - ☐ **Series-Series Topology: Example 04**
- ☐ **Basic MOSFET Amplifier Topologies**
- ☐ **Basic BJT Amplifier Topologies**

Practical Feedback Effect

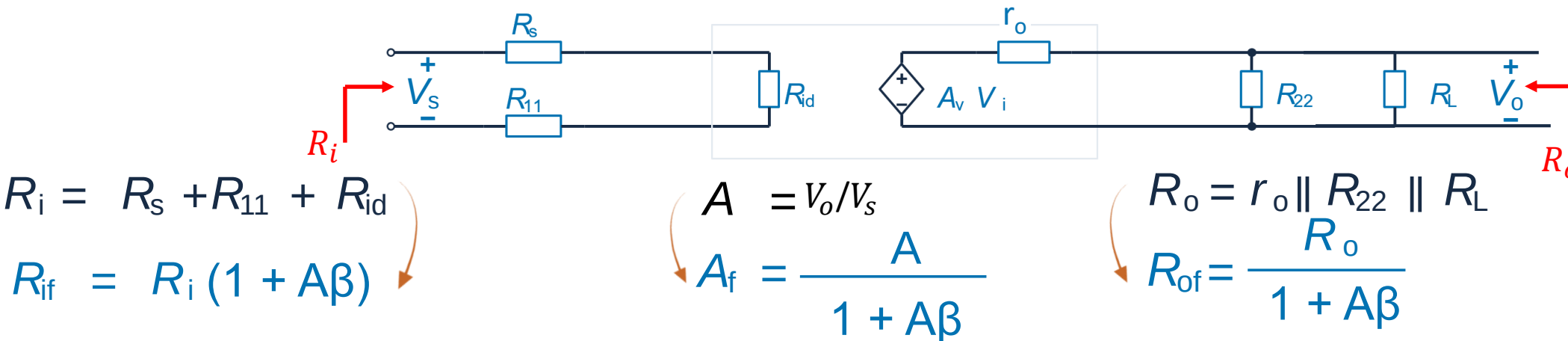


Practical Series-Shunt Feedback Analysis



- Feedback-network loading modifies the forward amplifier and its port resistances.
- R_{11} represents input loading. R_{22} represents output loading.

Forward amplifier with feedback-network loading



Input resistance excluding R_s

$$R_{if} = R_s + R_{in} \gg R_{in} = R_{if} - R_s$$

Output resistance excluding the load

$$R_{of} = R_{out} \parallel R_L \gg R_{out} = \left[\frac{1}{R_{of}} - \frac{1}{R_L} \right]^{-1}$$

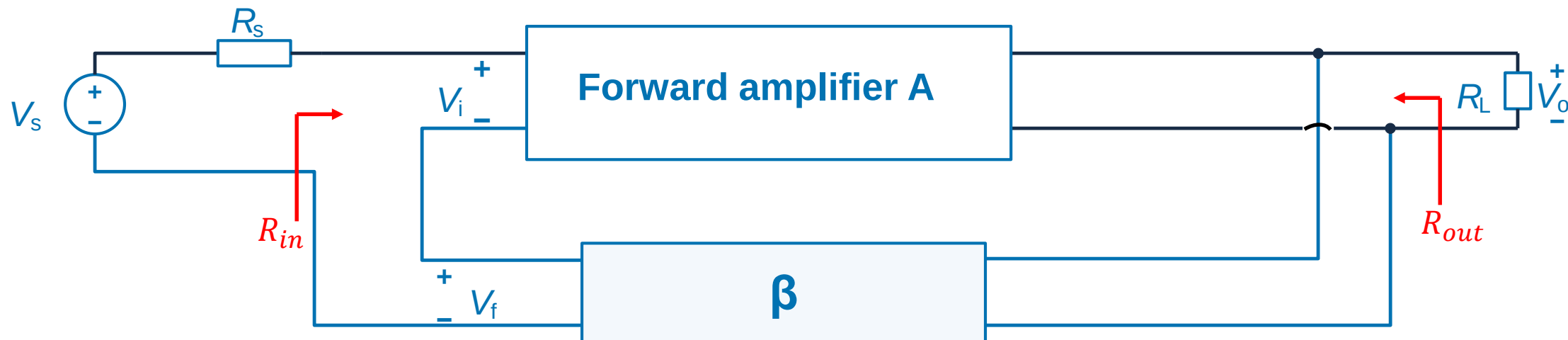


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Steps for Practical Feedback Analysis (1/4)



Step 1. Topology identification



Topology names list input mixing first and output sampling second.

Feedback topology	Input mixing	Output sampling
Series–series	Series	Series
Series–shunt	Series	Shunt
Shunt–series	Shunt	Series
Shunt–shunt	Shunt	Shunt

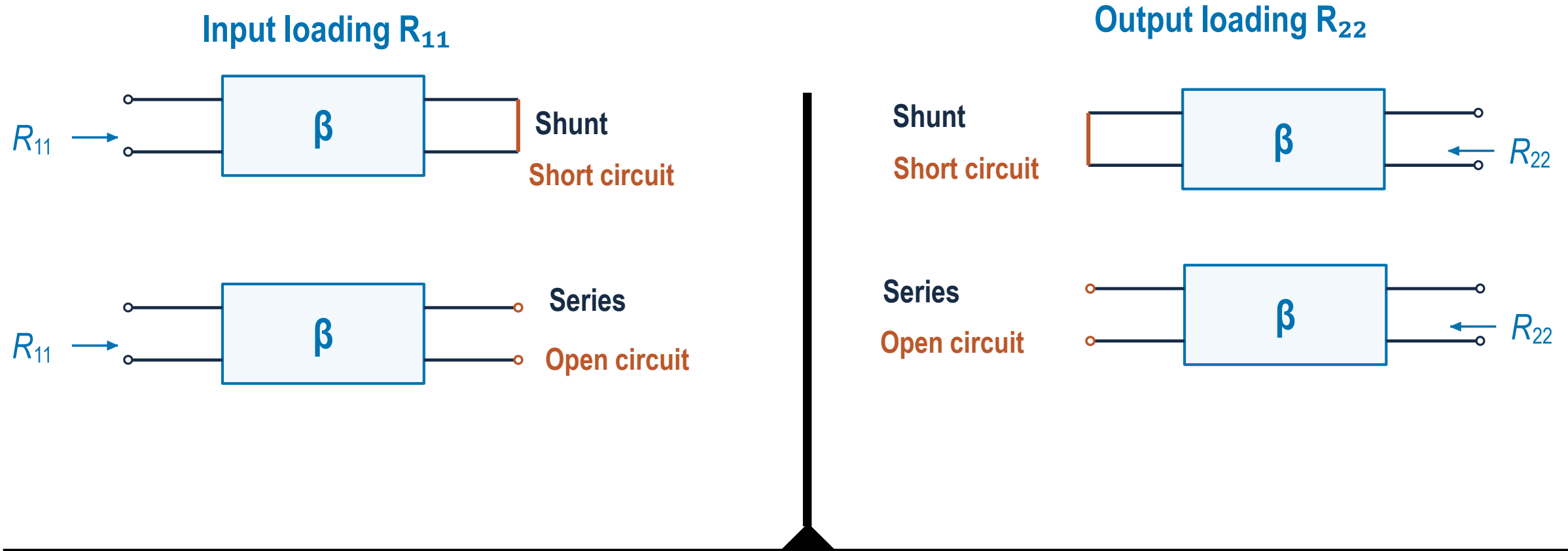


Steps for Practical Feedback Analysis (2/4)



Step 2. Feedback-Network Loading: R_{11} and R_{22}

- ❑ To get R_{11} and R_{22} → Measure the resistance at the port of interest after **terminating/destroy** the opposite port.



Opposite-port termination: **shunt** connection uses a **short**.
Series connection uses an **open**.



Steps for Practical Feedback Analysis (3/4)



Step 3. Feedback transfer ratio β



Note

Voltage feedback

Open the feedback output port



Current feedback

Short the feedback output port

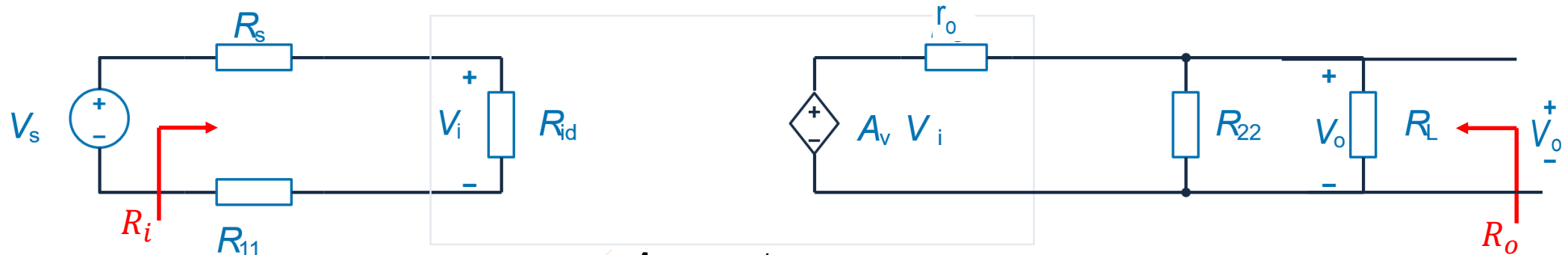


Steps for Practical Feedback Analysis (4/4)



Step 4: redraw the forward amplifier with the loading of the feedback network.

Step 5: Calculate the R_i , R_o , A



$$R_i = R_s + R_{11} + R_{id}$$

$$R_{if} = R_i (1 + A\beta)$$

$$A = V_o/V_s$$

$$A_f = \frac{A}{1 + A\beta}$$

$$R_o = r_o \parallel R_{22} \parallel R_L$$

$$R_{of} = \frac{R_o}{1 + A\beta}$$

Step 6: remove source and load contributions using the corresponding port connection.



Series connection

$$R_{in} = R_{if} - R_s$$

Shunt connection

$$R_{in} = \left[\frac{1}{R_{if}} - \frac{1}{R_s} \right]^{-1}$$

Series connection

$$R_{out} = R_{of} - R_L$$

Shunt connection

$$R_{out} = \left[\frac{1}{R_{of}} - \frac{1}{R_L} \right]^{-1}$$

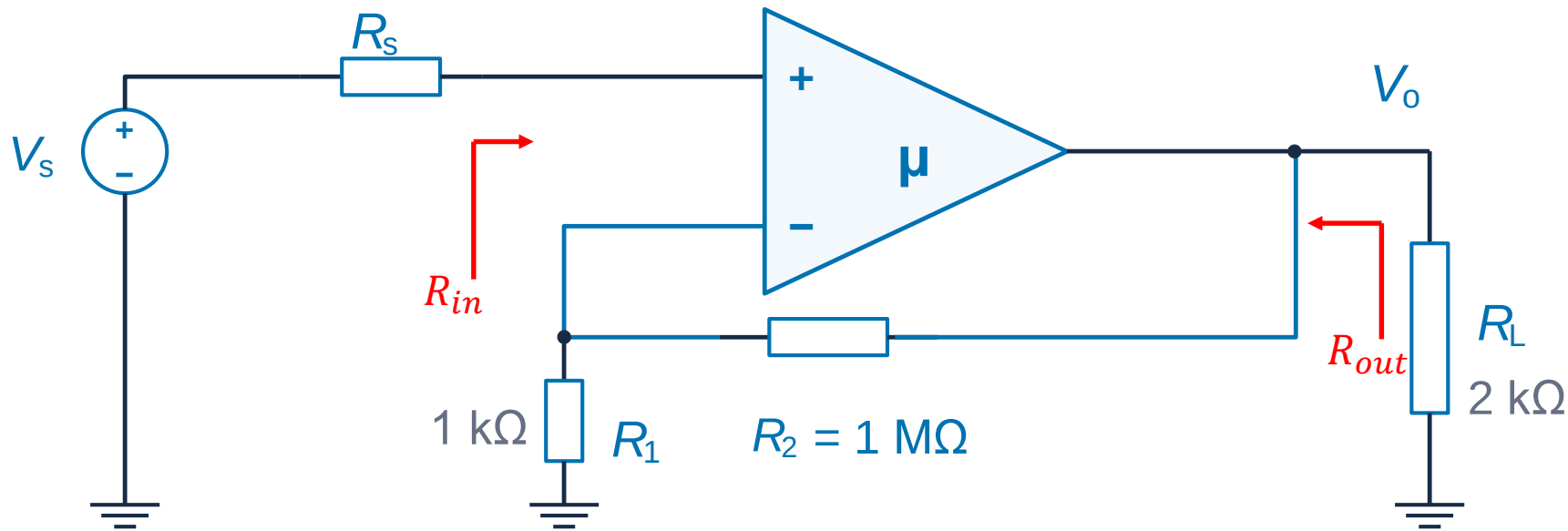


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Series-Shunt Topology: Example 01



Given circuit



Amplifier parameters

$$\mu = A_{v0} = 10^4$$

$$R_s = 10 \text{ k}\Omega$$

$$R_{id} = 100 \text{ k}\Omega$$

$$r_o = 1 \text{ k}\Omega$$

Required quantities

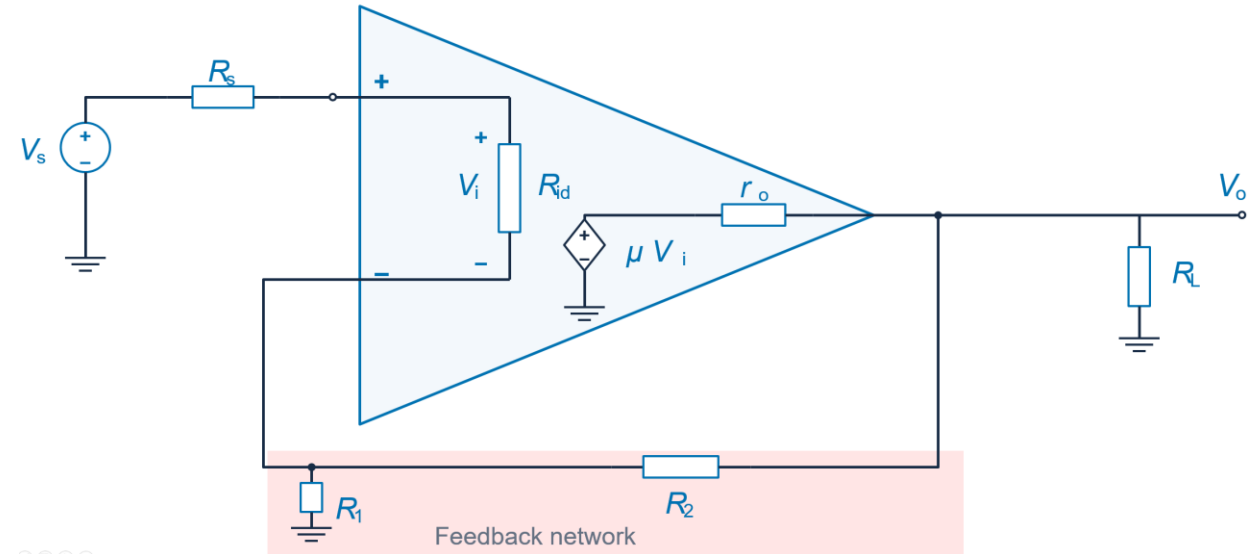
$$R_{in}, \quad R_{out}, \quad A_f = \frac{V_o}{V_s}$$

Series-Shunt Topology: Example 01



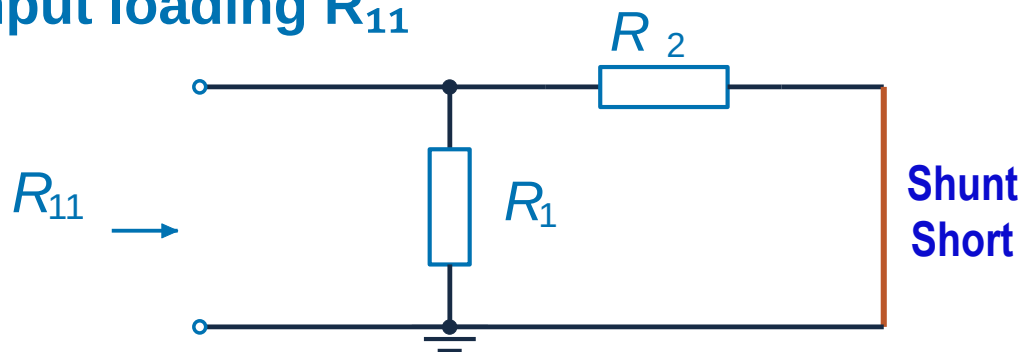
Step 1. Topology identification

Series – Shunt
(V) (V)



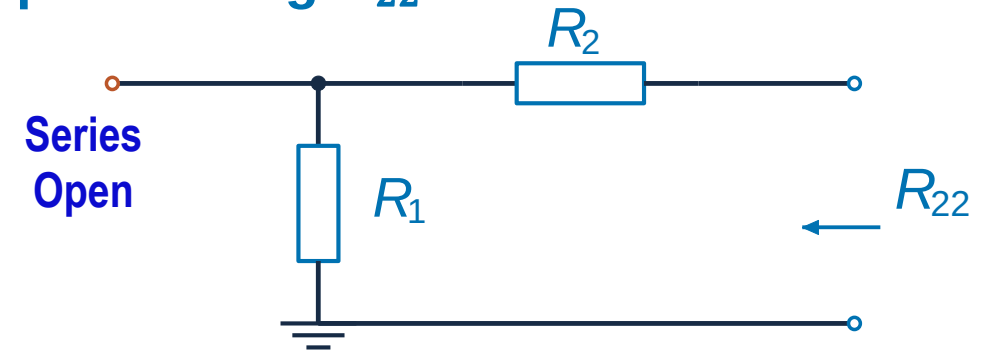
Step 2. Feedback-Network Loading: R_{11} and R_{22}

Input loading R_{11}



$$R_{11} = R_1 \parallel R_2 \\ = 999.0 \, \Omega \approx 1 \, \text{k}\Omega$$

Output loading R_{22}

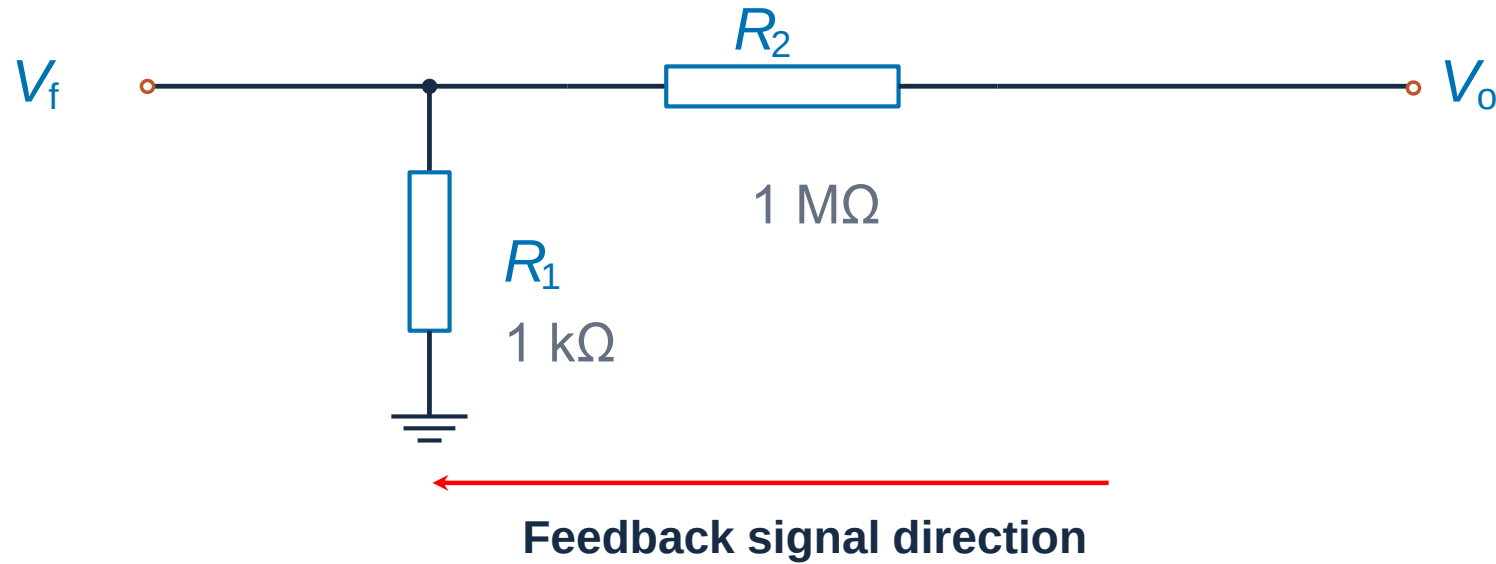


$$R_{22} = R_1 + R_2 \\ = 1.001 \, \text{M}\Omega \approx 1 \, \text{M}\Omega$$

Series-Shunt Topology: Example 01



Step 3. Feedback transfer ratio β



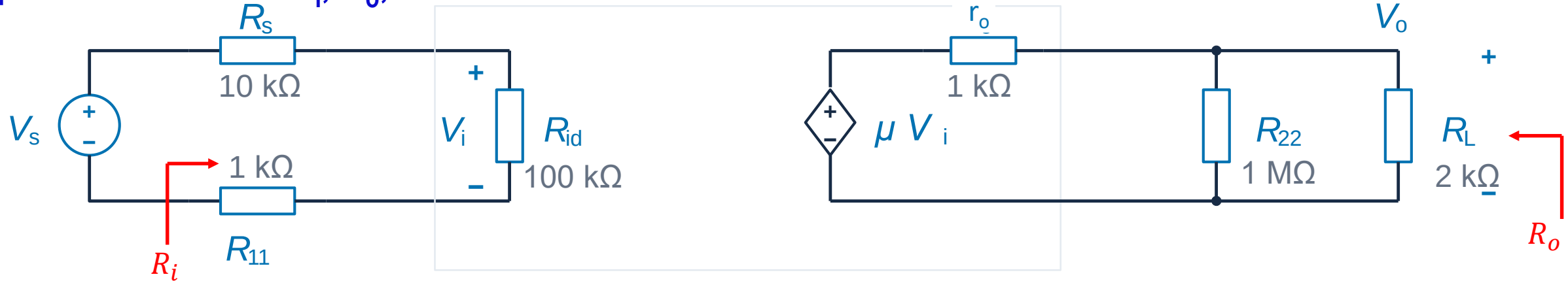
$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2} = \frac{1\text{ k}\Omega}{1\text{ k}\Omega + 1\text{ M}\Omega} \gg \beta = 9.990 \times 10^{-4}$$

Series-Shunt Topology: Example 01



Step 4: redraw the forward amplifier with the loading of the feedback network.

Step 5: Calculate the R_i , R_o , A



$$R_i = R_s + R_{11} + R_{id} = 111 \text{ k}\Omega$$

$$R_o = r_o \parallel R_{22} \parallel R_L = 666.22 \text{ }\Omega$$

$$A = \frac{V_o}{V_s} = \frac{V_o}{V_i} \times \frac{V_i}{V_s}$$

$$= \mu \frac{R_L \parallel R_{22}}{(R_L \parallel R_{22}) + r_o} \times \frac{R_{id}}{R_{id} + R_s + R_{11}}$$



$$A = 10^4 \times 0.6662 \times 0.9009 \approx 6000 \text{ V/V}$$

Series-Shunt Topology: Example 01



$$A = 6000 \quad \beta = 9.99 \times 10^{-4} \quad R_i = 111 \text{ k}\Omega \quad R_o = 666.2 \text{ }\Omega$$

Closed-loop gain

$$A_f = \frac{A}{1 + A\beta} = \frac{6000}{6.994} \approx 857.88$$

Input resistance with feedback

$$\begin{aligned} R_{if} &= R_i (1 + A\beta) \\ &= 111 \text{ k}\Omega \times 6.994 \\ &\approx 776.33 \text{ k}\Omega \end{aligned}$$

As the topology
Series -Shunt



Output resistance with feedback

$$\begin{aligned} R_{of} &= \frac{R_o}{1 + A\beta} \\ &= 666.2 / 6.994 \\ &\approx 95.25 \text{ }\Omega \end{aligned}$$

Series-Shunt Topology: Example 01



Step 6: remove source and load contributions using the corresponding port connection.

Input port: **series** mixing

$$R_{if} = R_{in} + R_s$$

$$R_{in} = R_{if} - R_s$$

$$\approx 776.33 \text{ k}\Omega - 10 \text{ k}\Omega$$

$$R_{in} \approx 766.33 \text{ k}\Omega$$

Output port: **shunt** sampling

$$R_{of} = R_{out} \parallel R_L$$

$$R_{out} = \left[\frac{1}{R_{of}} - \frac{1}{R_L} \right]^{-1}$$

$$R_{out} = \left[\frac{1}{95.25 \text{ }\Omega} - \frac{1}{2 \text{ k}\Omega} \right]^{-1}$$

$$R_{out} \approx 100 \text{ }\Omega$$

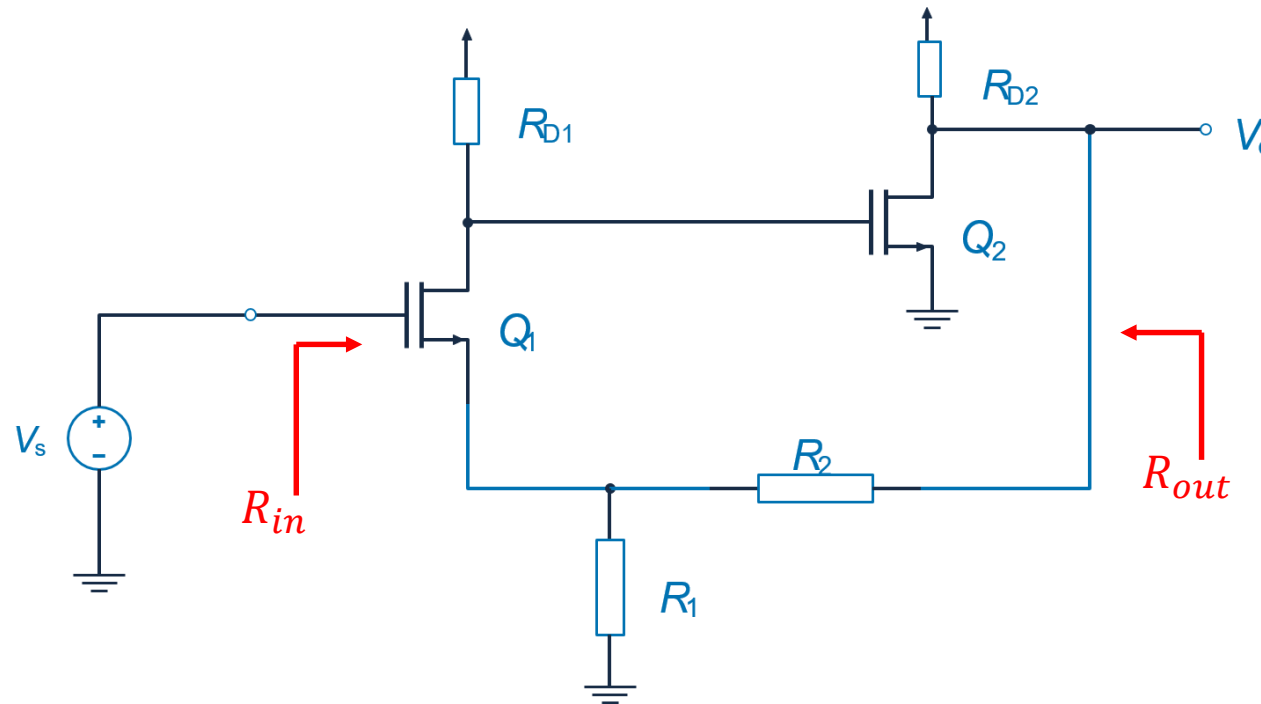


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Series-Shunt Topology: Example 02



Given circuit



Two-Transistor Feedback Amplifier

$$g_{m1} = g_{m2} = 4 \text{ mA/V}$$

$$R_{D1} = R_{D2} = 10 \text{ k}\Omega$$

$$R_1 = 1 \text{ k}\Omega$$

$$R_2 = 9 \text{ k}\Omega$$

Neglect transistor r_o

$$r_{o1}, r_{o2} \rightarrow \infty$$

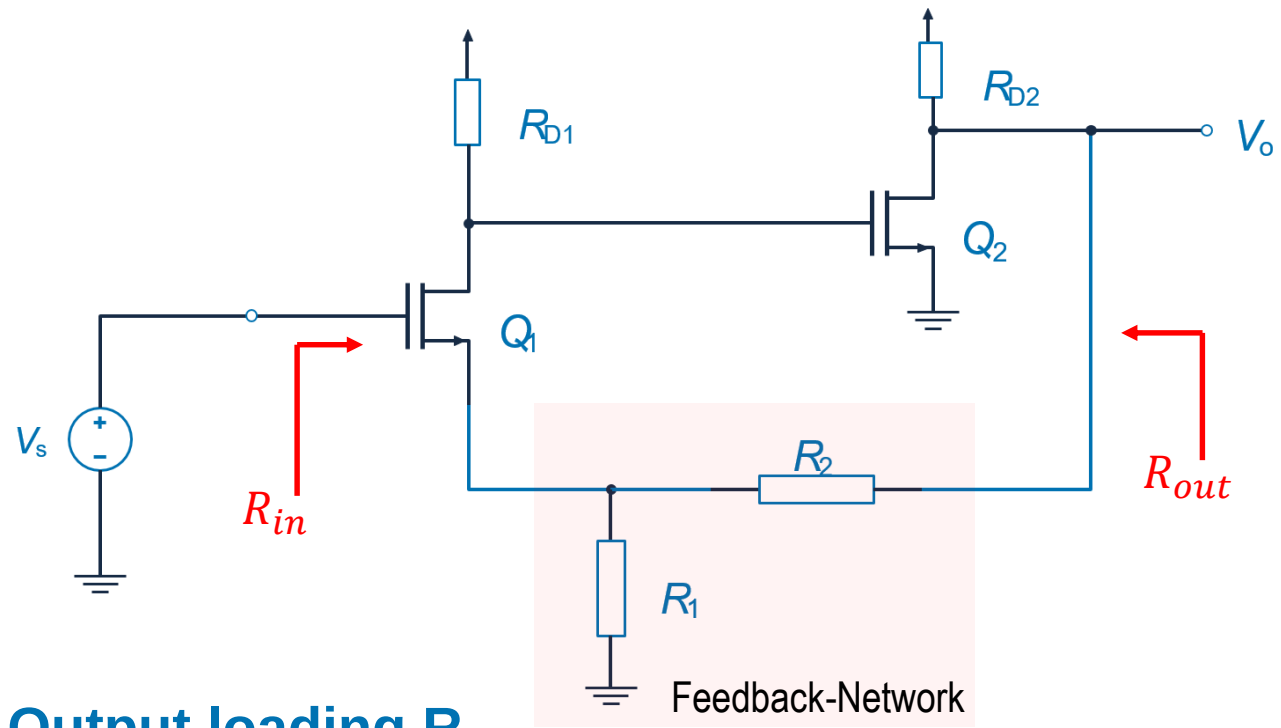
Required quantities

$$R_{in}, R_{out}, A_f = \frac{V_o}{V_s}$$

Series-Shunt Topology: Example 02

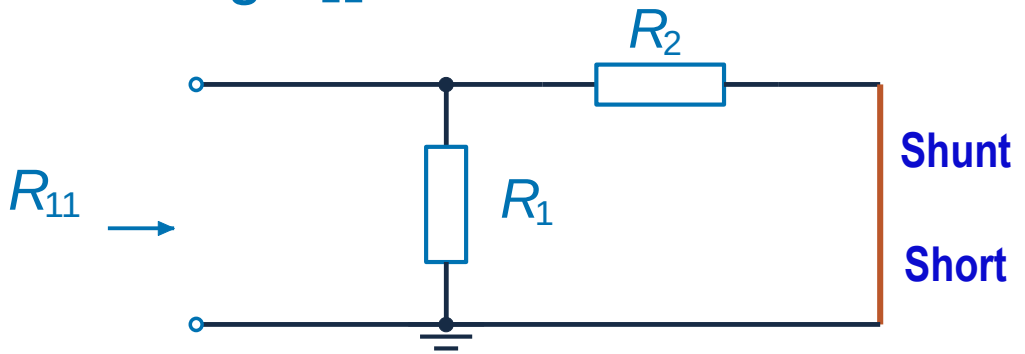
Step 1. Topology identification

Series – Shunt
(V) (V)



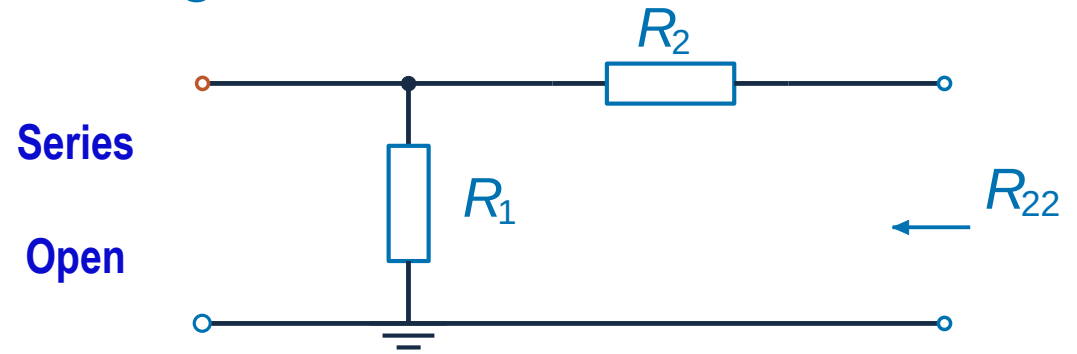
Step 2. Feedback-Network Loading: R_{11} and R_{22}

Input loading R_{11}



$$R_{11} = R_1 \parallel R_2 = 0.9 \text{ k}\Omega$$

Output loading R_{22}



$$R_{22} = R_1 + R_2 = 10 \text{ k}\Omega$$

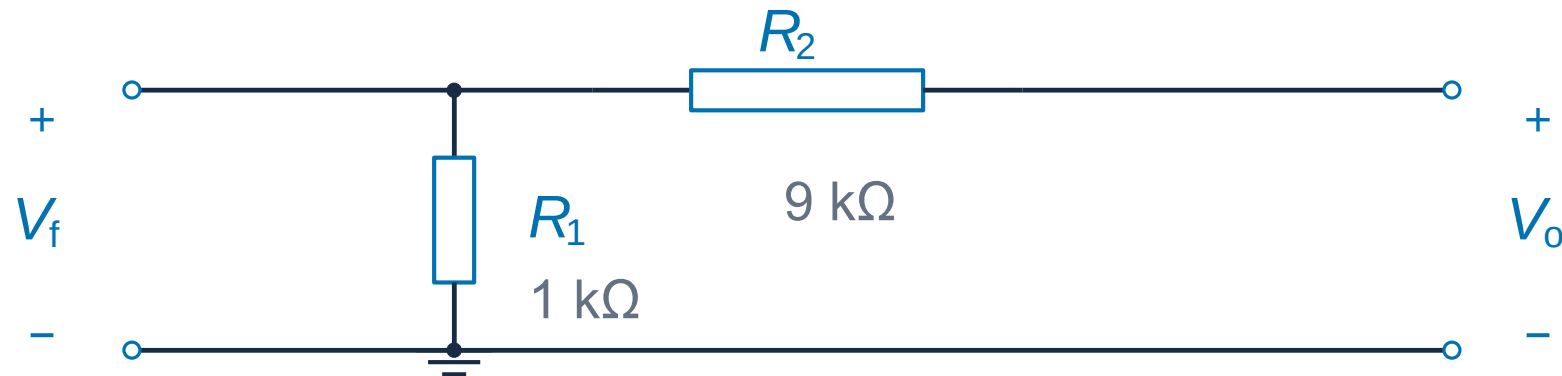
Series-Shunt Topology: Example 02



Step 3. Feedback transfer ratio β

Port 1: open voltage-mixing port

Port 2: applied output voltage



$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2} = \frac{1 \text{ k}\Omega}{1 \text{ k}\Omega + 9 \text{ k}\Omega} = 0.1$$

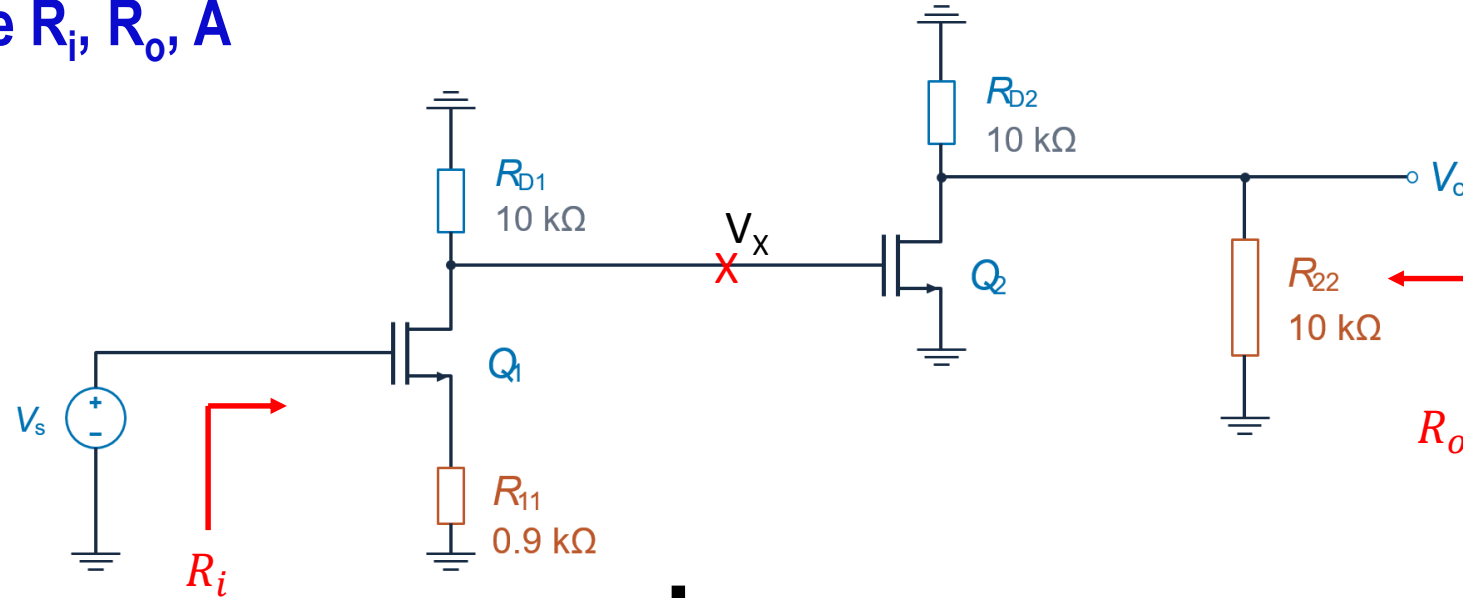
The feedback network returns 10% of the output voltage.

Series-Shunt Topology: Example 02



Step 4: redraw the forward amplifier with the loading of the feedback network.

Step 5: Calculate the R_i , R_o , A



$I_i = 0$ (MOS gate current)

$$R_i = \frac{V_s}{I_i} \rightarrow \infty$$

$$R_o = R_{D2} \parallel R_{22} = 10 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 5 \text{ k}\Omega$$

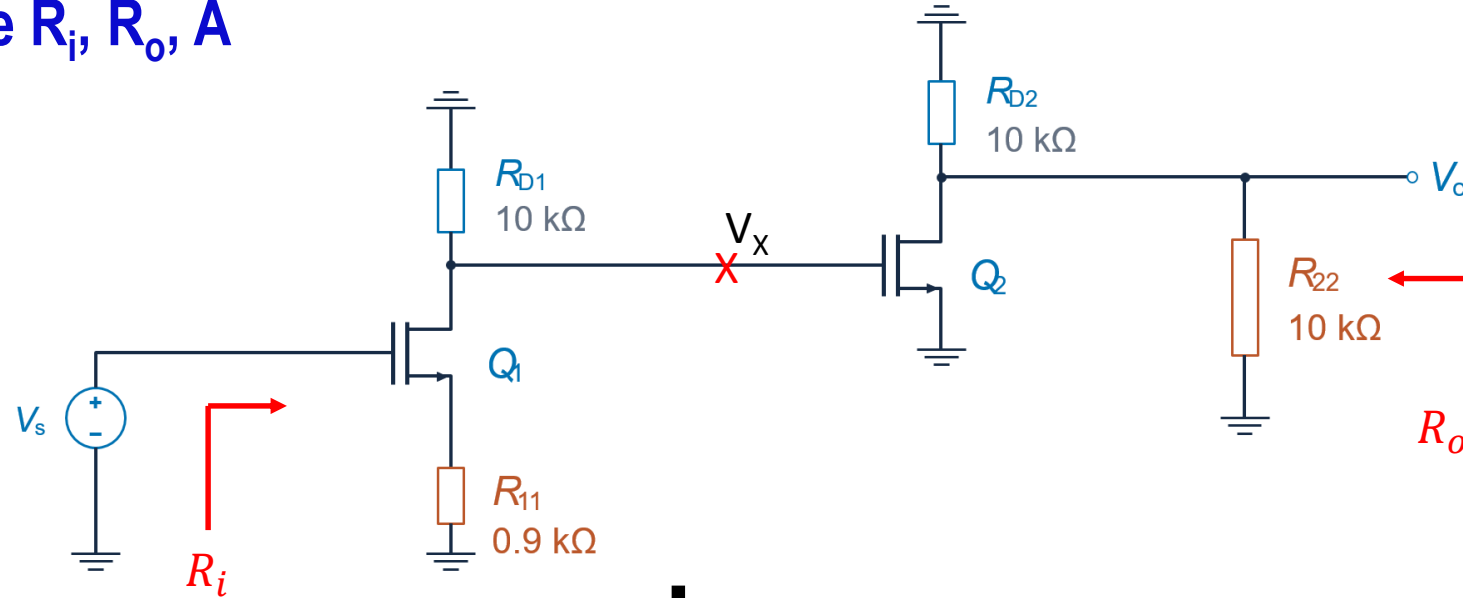
Series-Shunt Topology: Example 02



Step 4: redraw the forward amplifier with the loading of the feedback network.

Step 5: Calculate the R_i , R_o , A

$$\therefore A = \frac{V_o}{V_s} = \frac{V_o}{V_x} \cdot \frac{V_x}{V_s}$$



Stage 1: source degeneration

$$A_1 = V_x / V_s$$

$$A_1 = \frac{-g_{m1} R_{D1}}{1 + g_{m1} R_{11}} \approx -8.696 \text{ V/V}$$

Stage 2: loaded common source

$$A_2 = V_o / V_x$$

$$A_2 = -g_{m2} (R_{D2} \parallel R_{22}) = -20 \text{ V/V}$$

Overall loaded forward gain

$$A = A_1 \times A_2 = (40 / 4.6) \times 20 \approx \mathbf{+173.91 \text{ V/V}}$$

Series-Shunt Topology: Example 02



Closed-Loop Gain and Port Resistances

$A \approx 173.913 \text{ V/V}$, $\beta = 0.1$, $R_i \rightarrow \infty$ and $R_o = 5 \text{ k}\Omega$.

Closed-loop voltage gain

$$A_f = \frac{A}{1 + A\beta} = \frac{173.913}{18.3913} \approx 9.456 \text{ V/V}$$

Input resistance with feedback

$$R_{if} = R_i (1 + A\beta)$$

$R_i \rightarrow \infty$ in the ideal MOS model

$$R_{if} \rightarrow \infty$$

Gate leakage and transistor output conductance are neglected.

Output resistance with feedback

$$\begin{aligned} R_{of} &= \frac{R_o}{1 + A\beta} \\ &= \frac{5000 \Omega}{18.3913} \approx 271.87 \Omega \end{aligned}$$

Series-Shunt Topology: Example 02



Step 6: remove source and load contributions using the corresponding port connection.

Input port: series connection

$$R_{if} = R_{in} + R_s$$

$$R_{in} = R_{if} - R_s$$

This circuit: ideal voltage source

$$R_s = 0 \, \Omega$$

$$R_{in} = R_{if} - 0$$

$$R_{in} \rightarrow \infty$$

Output port: shunt connection

$$R_{of} = R_{out} \parallel R_L$$

$$R_{out} = \left[\frac{1}{R_{of}} - \frac{1}{R_L} \right]^{-1}$$

This circuit: no external load

$$R_L \rightarrow \infty$$

$$R_{out} = R_{of}$$

$$R_{out} \approx 271.87 \, \Omega$$

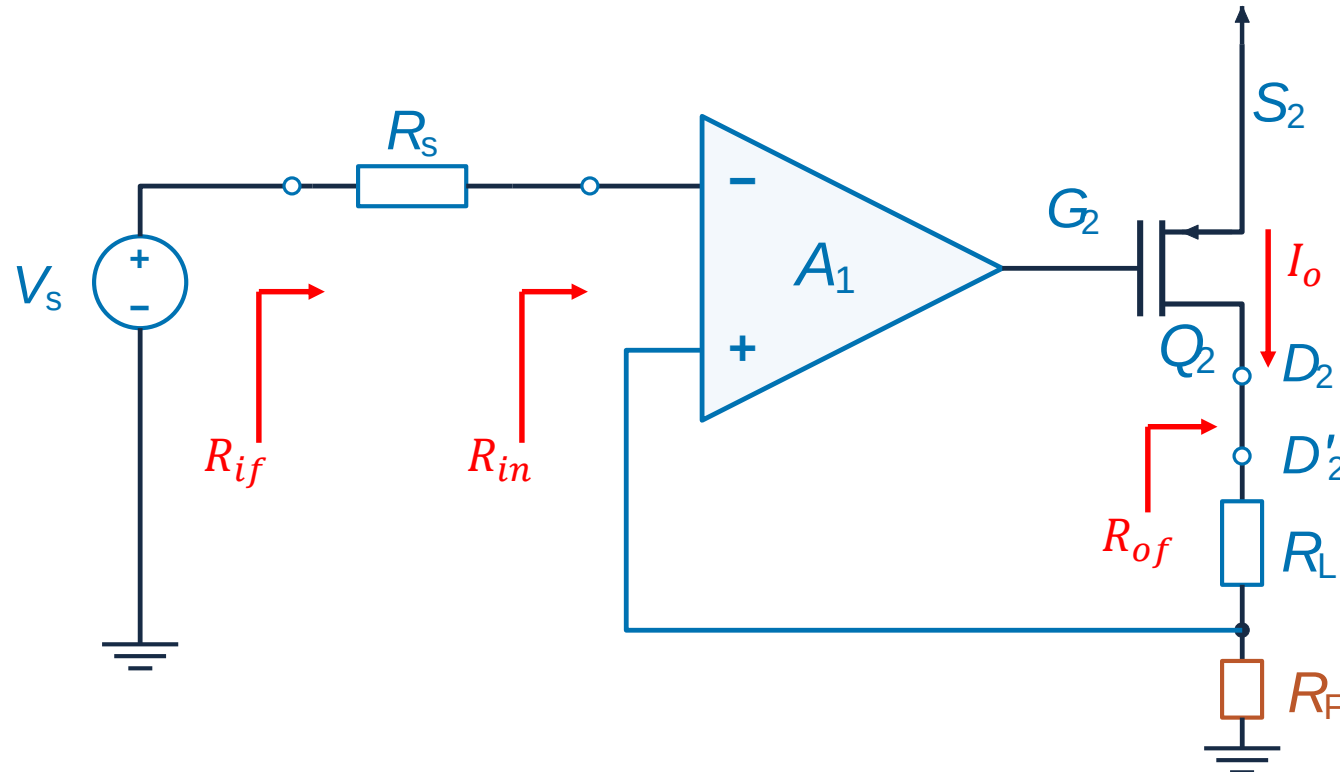


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Series-Series Topology: Example 03



Given circuit



Given parameters

Differential amplifier

A_1 , R_{id} , R_{o1}

MOSFET Q_2

g_{m2} , r_{o2}

Circuit resistances

R_s , R_F , R_L

Output resistance R_{out} is measured between the load terminals, looking into the output loop.

Required quantities

$$R_{in}, R_{out}, A_f = \frac{I_o}{V_s}$$

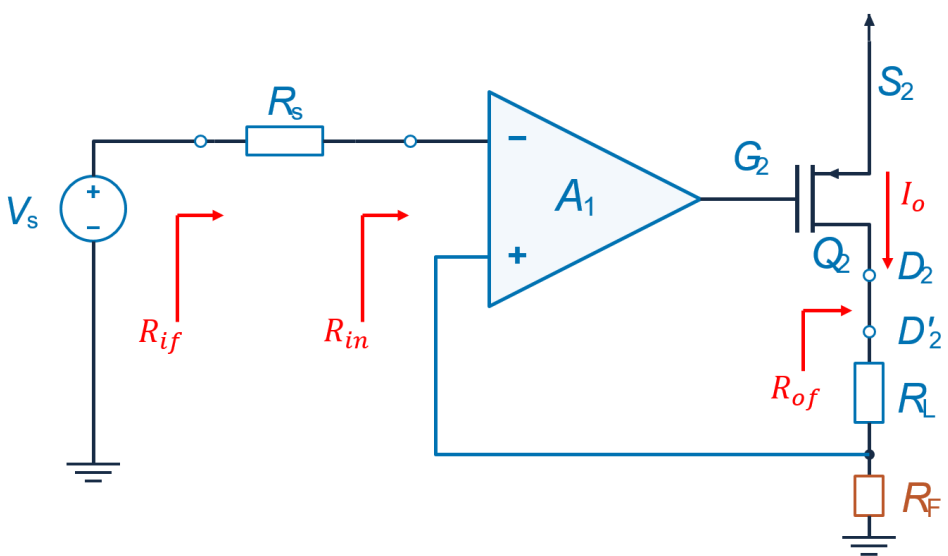
Series-Series Topology: Example 03

Step 1. Topology identification

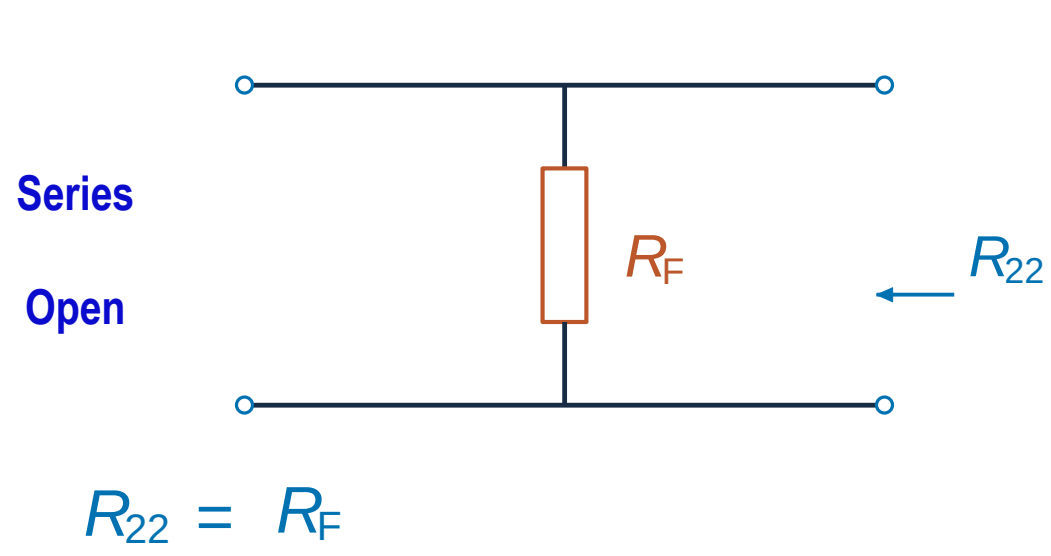
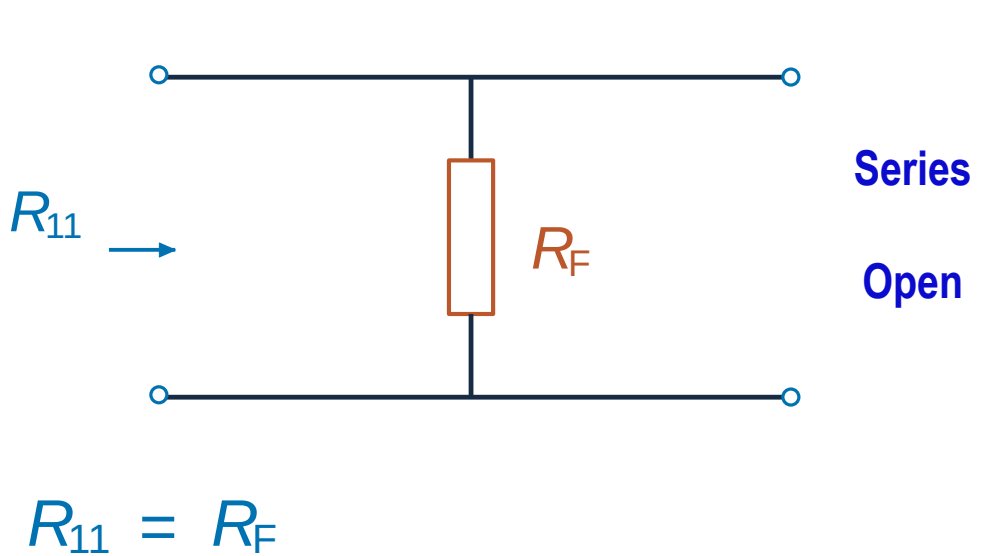
Series –Series

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Step 2. Feedback-Network Loading: R_{11} and R_{22}



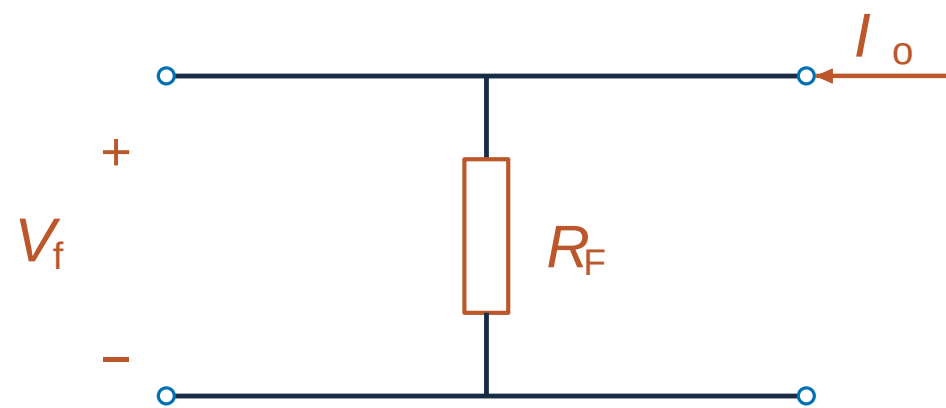
Series-Series Topology: Example 03



Step 3. Feedback transfer ratio β

Port 1: open voltage-mixing port

Port 2: applied output Current



$$V_f = I_o R_F \gg \gg \beta = \frac{V_f}{I_o} = R_F$$

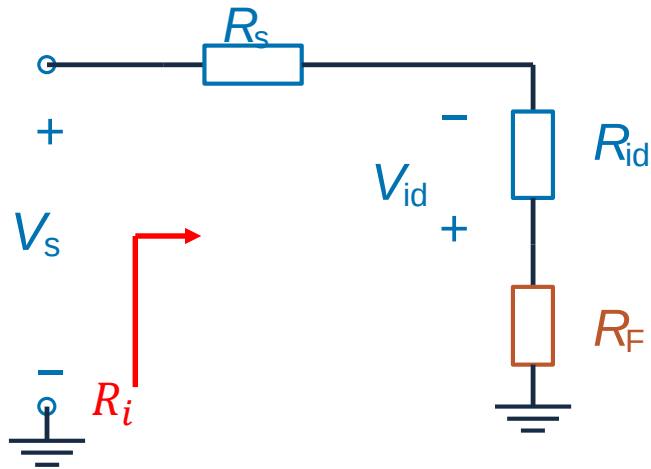
Series-Series Topology: Example 03



Step 4: redraw the forward amplifier with the loading of the feedback network.

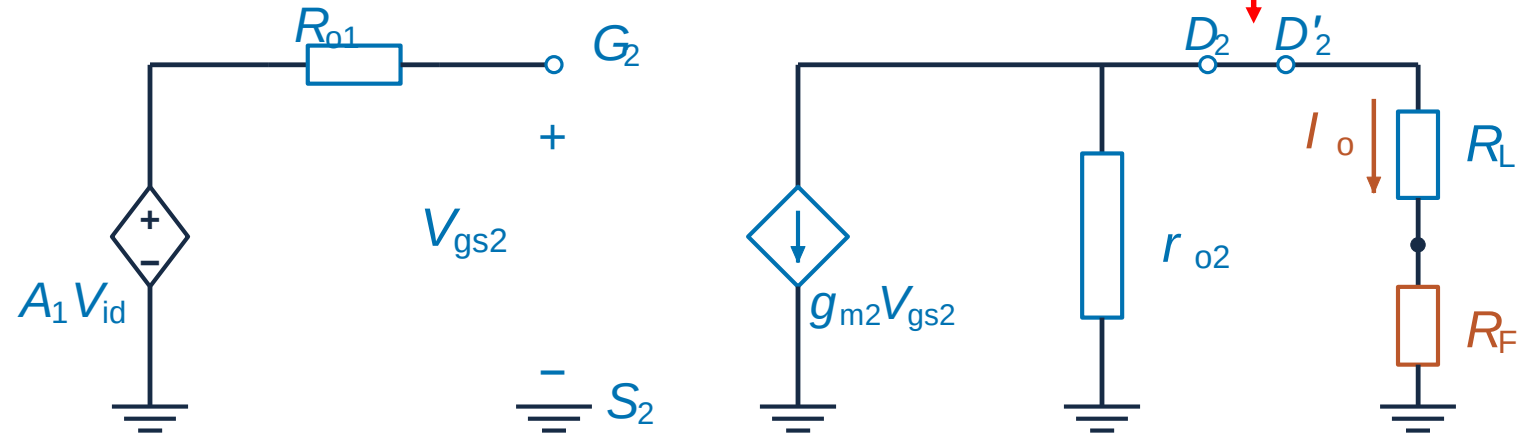
Step 5: Calculate the R_i , R_o , A

Differential stage



$$R_i = R_s + R_{id} + R_F$$

PMOS output loop



$$R_o = r_{o2} + R_L + R_F$$

Output test: set $V_s = 0$
and break the loop
between D_2 and D'_2 .

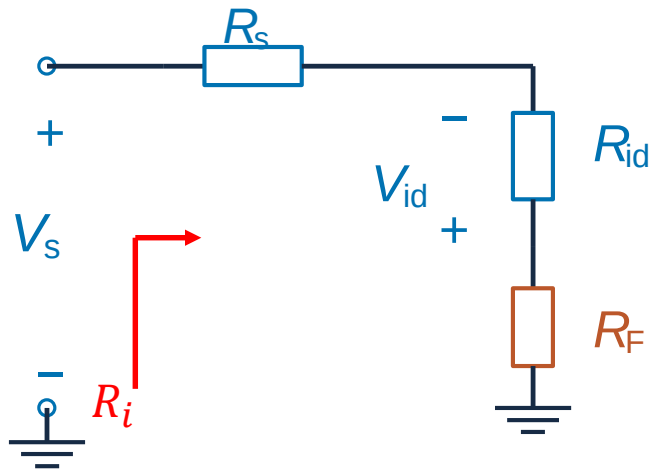
Series-Series Topology: Example 03



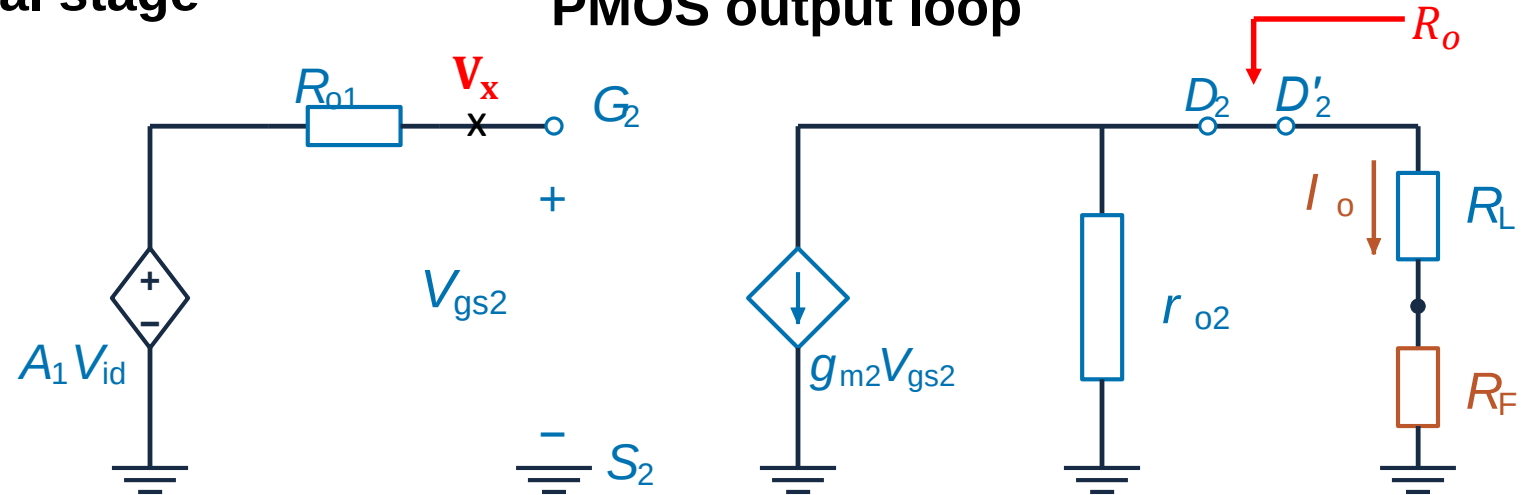
Step 4: redraw the forward amplifier with the loading of the feedback network.

Step 5: Calculate the R_i , R_o , A

Differential stage



PMOS output loop



$$\therefore A = \frac{I_o}{V_s} = \frac{I_o}{V_x} \cdot \frac{V_x}{V_s}$$



$$A = \frac{A_1 R_{id}}{R_{id} + R_s + R_F} \cdot \frac{g_{m2} r_{o2}}{r_{o2} + R_L + R_F}$$

$$V_{gs2} = V_x = A_1 V_{id}$$

$$V_{id} = -V_s \frac{R_{id}}{R_{id} + R_s + R_F}$$



$$\boxed{\frac{V_x}{V_s}} = A_1 \frac{R_{id}}{R_{id} + R_s + R_F}$$

$$I_o = -g_{m2} V_{gs2} \frac{r_{o2}}{r_{o2} + R_L + R_F}$$

$$V_{gs2} = V_x$$

$$\boxed{\frac{I_o}{V_x}} = -g_{m2} \frac{r_{o2}}{r_{o2} + R_L + R_F}$$



Series-Series Topology: Example 03



Closed-Loop Gain and Port Resistances

Closed-loop voltage gain

$$A_f = \frac{A}{1 + A\beta}$$

Input resistance with feedback

$$R_{if} = R_i (1 + A\beta)$$

Output resistance with feedback

$$R_{of} = R_o (1 + A\beta)$$

Series-Series Topology: Example 03



Step 6: remove source and load contributions using the corresponding port connection.

Input resistance

$$R_{in} = R_{if} - R_s$$

Output resistance

$$R_{out} = R_{of} - R_L$$

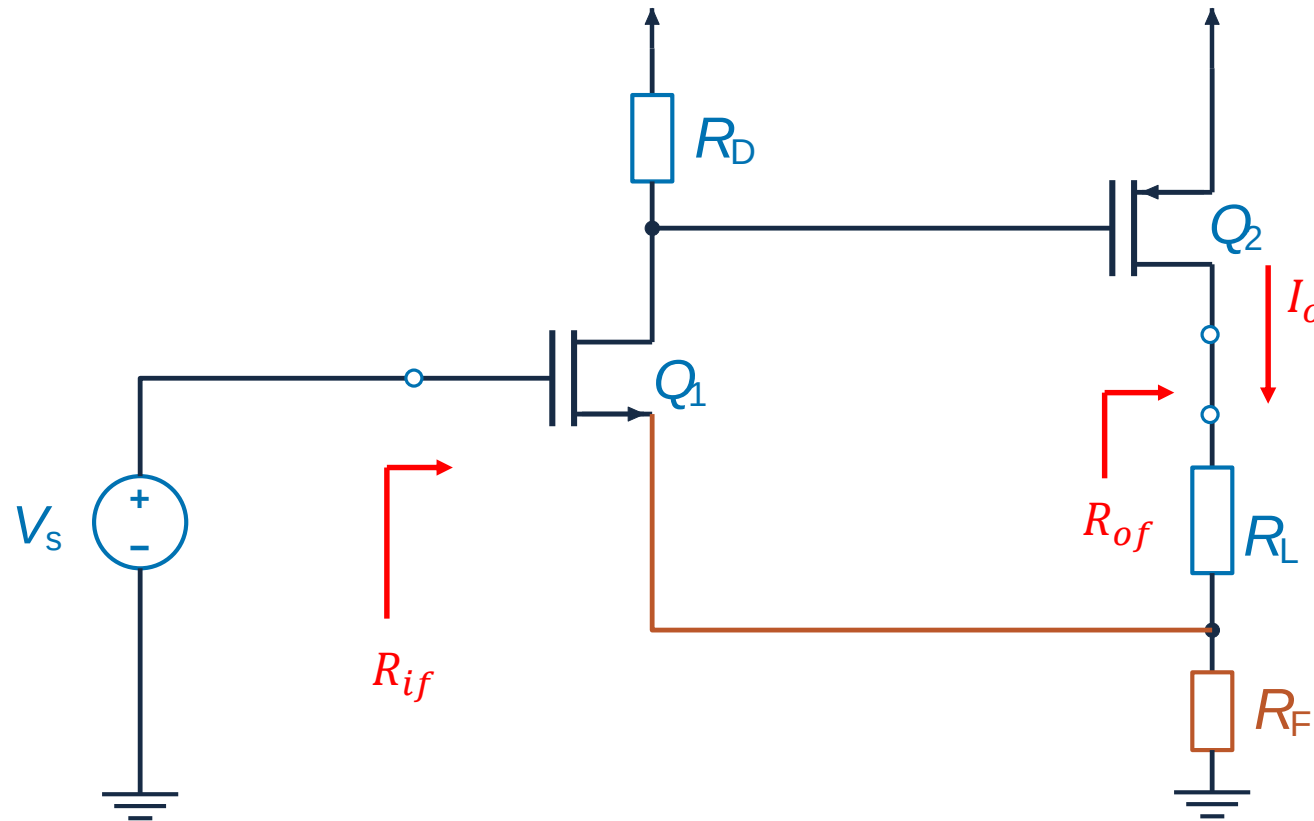


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Series-Series Topology: Example 04



Given circuit



Given parameters

$$g_{m1} = g_{m2} = 4 \text{ mA/V}$$

$$R_D = 20 \text{ k}\Omega$$

$$r_{o2} = 20 \text{ k}\Omega$$

$$R_F = 100 \Omega$$

$$R_L = 1 \text{ k}\Omega$$

Neglect r_{o1} . Use r_{o2} only for output resistances.

Required quantities

$$R_{in}, R_{out}, A_f = \frac{I_o}{V_s}$$

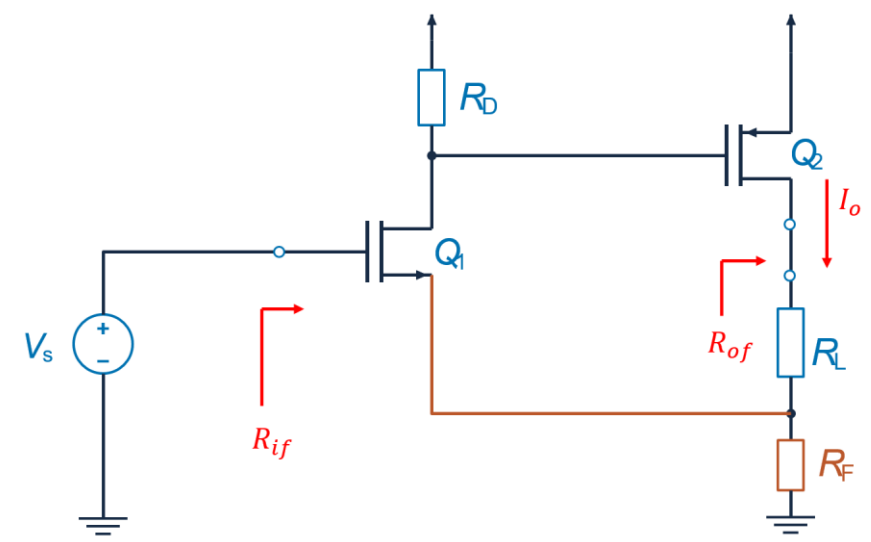
Series-Series Topology: Example 04

Step 1. Topology identification

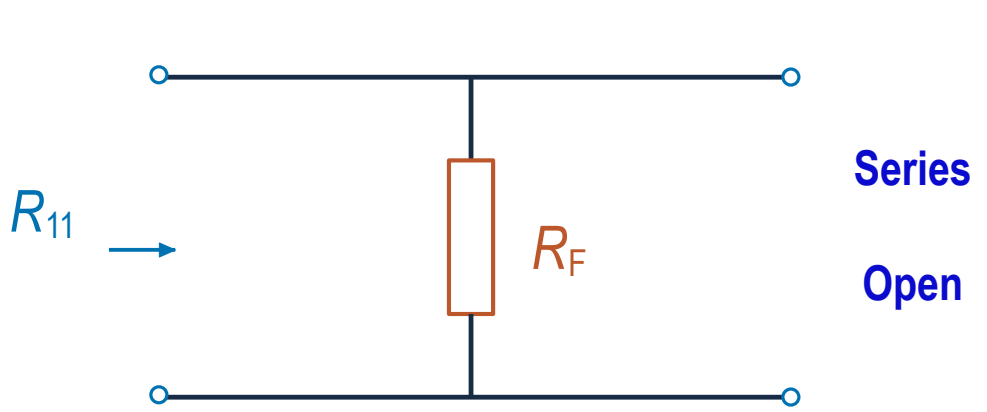
Series –Series

(v)

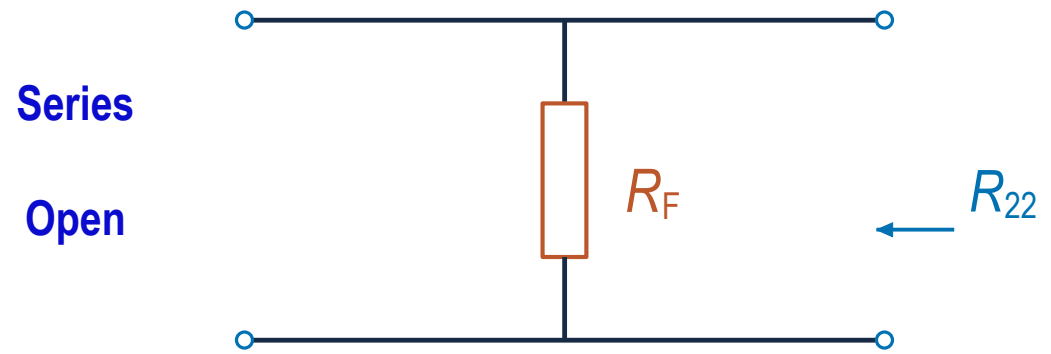
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Step 2. Feedback-Network Loading: R_{11} and R_{22}



$$R_{11} = R_F = 100 \, \Omega$$



$$R_{22} = R_F = 100 \, \Omega$$

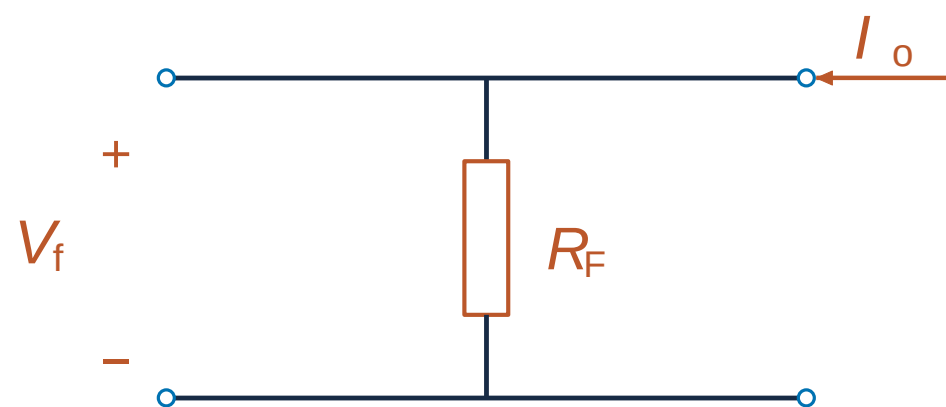
Series-Series Topology: Example 04



Step 3. Feedback transfer ratio β

Port 1: open voltage-mixing port

Port 2: applied output Current



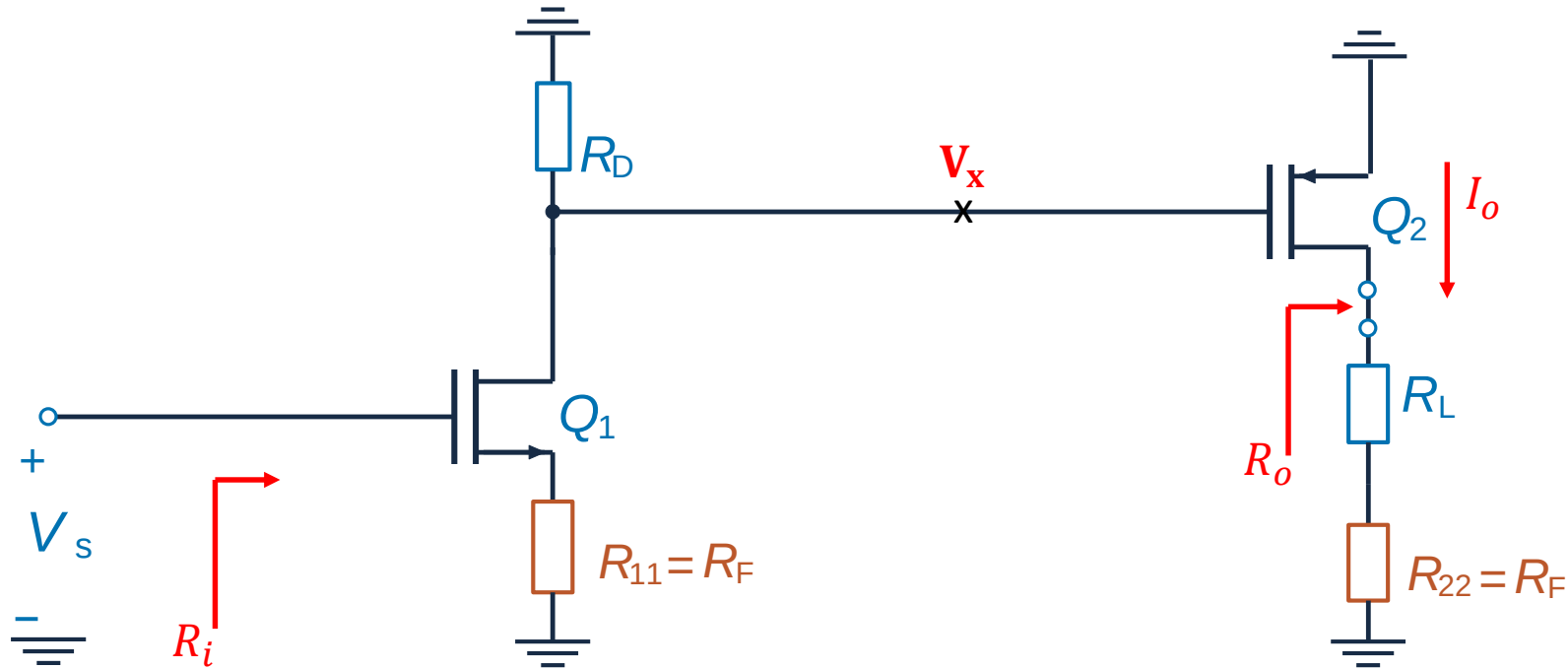
$$V_f = I_o R_F \gg \gg \beta = \frac{V_f}{I_o} = R_F \gg \gg \beta = 100 \, \Omega$$

Series-Series Topology: Example 04



Step 4: redraw the forward amplifier with the loading of the feedback network.

Step 5: Calculate the R_i , R_o , A



$I_i = 0$ (MOS gate current)

$$R_i = \frac{V_s}{I_i} \rightarrow \infty$$

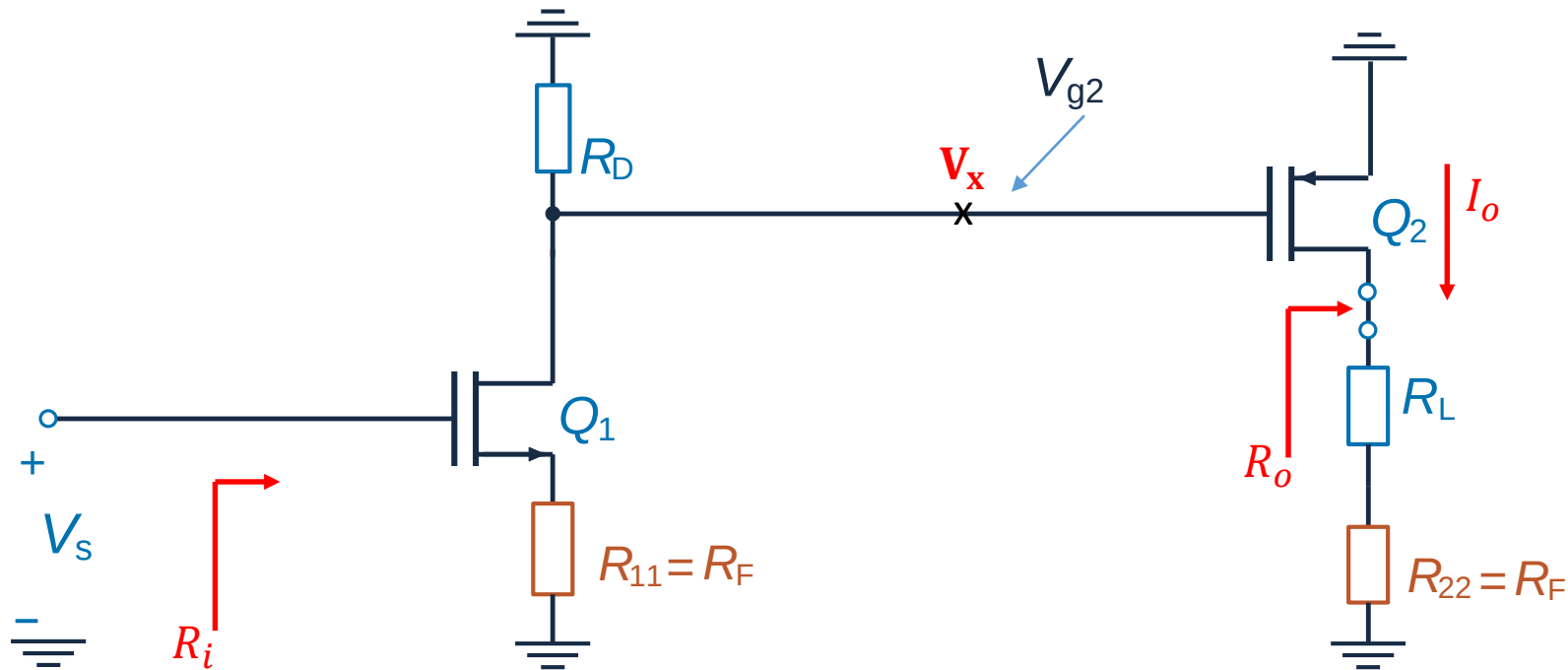
$$R_o = r_{o2} + R_L + R_F = 21.1 \text{ k}\Omega$$

Series-Series Topology: Example 04



Step 4: redraw the forward amplifier with the loading of the feedback network.

Step 5: Calculate the R_i , R_o , A



$$\therefore A = \frac{I_o}{V_s} = \frac{I_o}{V_x} \cdot \frac{V_x}{V_s} \ggggg A = \frac{g_{m1}g_{m2}R_D}{1 + g_{m1}R_F} = 0.22857 \text{ A/V}$$

$$V_x = -\frac{g_{m1}R_D}{1 + g_{m1}R_F} V_s \ggggg \boxed{\frac{V_x}{V_s}} = -\frac{g_{m1}R_D}{1 + g_{m1}R_F}$$

$$I_o = -g_{m2}V_{g2}$$

$$\boxed{\frac{I_o}{V_x}} = -g_{m2}$$

Series-Series Topology: Example 04



Closed-Loop Gain and Port Resistances

Closed-loop voltage gain

$$A_f = \frac{A}{1 + A\beta} = 9.58 \text{ mA/V}$$

Input resistance with feedback

$$R_{if} = R_i (1 + A\beta) \rightarrow \infty$$

Output resistance with feedback

$$R_{of} = R_o (1 + A\beta) = 503.4 \text{ k}\Omega$$

Series-Series Topology: Example 04



Step 6: remove source and load contributions using the corresponding port connection.

Input resistance

$$R_i = R_{if} = R_{in} = \infty$$

Ideal MOS gate and zero external source resistance

Output resistance

$$\begin{aligned} R_{out} &= R_{of} - R_L \\ &= 503.3857 \text{ k}\Omega - 1 \text{ k}\Omega = 502.4 \text{ k}\Omega \end{aligned}$$



- ❑ **Practical Feedback Effect**
- ❑ **Steps for Practical Feedback Analysis**
 - ❑ **Series-Shunt Topology: Example 01**
 - ❑ **Series-Shunt Topology: Example 02**
 - ❑ **Series-Series Topology: Example 03**
 - ❑ **Series-Series Topology: Example 04**
- ❑ **Basic MOSFET Amplifier Topologies**
- ❑ **Basic BJT Amplifier Topologies**



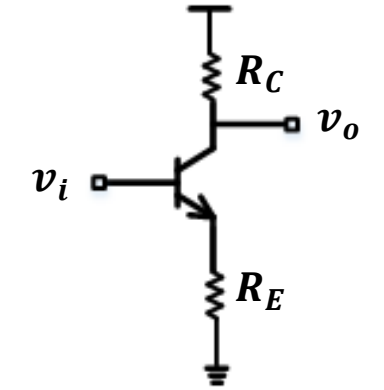
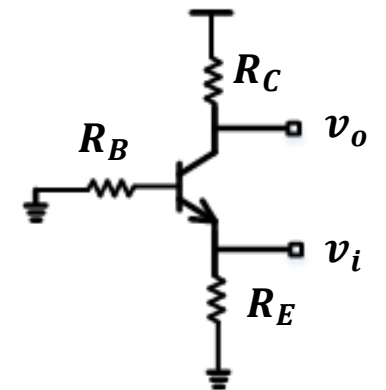
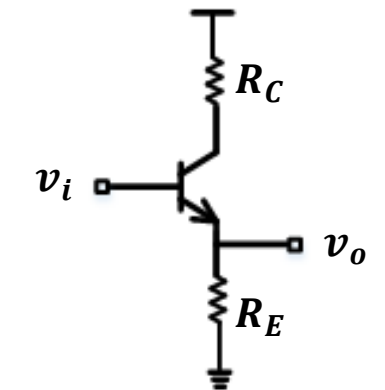
Basic MOSFET Amplifier Topologies

	Common Source	Common Gate	Common Drain
R_{in}	∞	$R_S \parallel \frac{1}{g_m}$	∞
R_{out}	R_D	R_D	$R_S \parallel \frac{1}{g_m} = \frac{R_S}{1 + g_m R_S}$
G_m	$\frac{-g_m}{1 + g_m R_S}$	g_m	g_m
A_v	$\frac{-g_m R_D}{1 + g_m R_S}$	$g_m R_D$	$\frac{g_m R_S}{1 + g_m R_S}$



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Basic BJT Amplifier Topologies

	Common Emitter	Common Base	Common Collector
			
R_{in}	$r_{\pi}[1 + g_m R_E]$	$R_E \parallel \frac{1}{g_m} \left[1 + \frac{R_B}{r_{\pi}} \right] \approx R_E \parallel \frac{1}{g_m}$ if $r_{\pi} \gg R_B$	$r_{\pi}[1 + g_m R_E]$
R_{out}	R_C	R_C	$R_E \parallel \frac{1}{g_m} = \frac{R_E}{1 + g_m R_E}$
G_m	$\frac{-g_m}{1 + g_m R_E}$	$\frac{g_m}{1 + \frac{R_B}{r_{\pi}}} \approx g_m$ if $r_{\pi} \gg R_B$	g_m
A_v	$\frac{-g_m R_C}{1 + g_m R_E}$	$\frac{g_m R_C}{1 + \frac{R_B}{r_{\pi}}} \approx g_m R_C$ if $r_{\pi} \gg R_B$	$\frac{g_m R_E}{1 + g_m R_E}$

Thanks!

